

students

CSV to PDF feedback reports

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Target of the program

The program converts CSV files exported from the Moodle learning management system (survey results for “Discussion Leading Feedback”) into individual PDF feedback reports for each discussion leader. It is designed for Czech-locale Moodle exports.

Columns are identified by pattern matching on the header names, making the program compatible with different Moodle export variants (16 or 17 columns, with or without the optional “Positive feedback” and “Group” columns).

Required columns

Keyword	Description
Leader name	Name of the discussion leader
Preparation	Score: leader was well-prepared (1–5)
Introduction	Score: clear topic introduction (1–5)
Inclusion	Score: everyone participated (1–5)
Fav question	Open text: most interesting question
What learned	Open text: one thing learned
Negative feedback	Open text: what could be improved
ID	Student identifier (for deduplication)

How to install

Option 1: Windows executable

Download students.exe from the [release page](#) and place it in a folder without spaces in the path. Add the folder to your PATH environment variable.

Option 2: Run from source

Requires Python 3.9 or later. Install the dependencies:

```
pip install -r requirements.txt
```

How to use

Navigate to the folder containing the CSV file and run:

```
students
```

or, when running from source:

```
python src/students/main.py
```

The program will:

- Ask for the CSV file name.
- Perform a structural analysis of the file (encoding, delimiter, column validation) and report any issues.
- Clean and normalize leader names to avoid duplicate PDFs.
- Generate one PDF per leader in a pdf/ subfolder.

File analysis

Before importing data, the program checks the CSV for encoding (UTF-8 with/without BOM, Windows-1250, Latin-1), delimiter (comma, semicolon, tab), missing or duplicate columns, non-numeric values in score columns, and empty or malformed rows. Issues are classified as ERROR (blocks execution), WARNING (may cause imprecise results), or INFO (informational).

Name cleaning

Three layers of normalization reduce duplicate PDF generation:

- Strip — removes whitespace and normalizes Unicode.
- Structural — merges surname-only entries with full names and unifies reversed name order.
- Fuzzy — catches typos and accent variants using Levenshtein distance. All corrections are logged to the terminal.

Testing

A test runner is included. From the project root:

```
bash test/run_tests.sh # run all tests
```

```
bash test/run_tests.sh clean # clean test output
```

Project structure

File	Description
src/students/main.py	Entry point
src/students/file_analysis.py	CSV structural validation
src/students/data_import.py	Column mapping and import
src/students/data_cleaning.py	Leader name normalization
src/students/pdfgen.py	PDF generation
src/students/styles.py	Garamond paragraph styles
src/fonts/	Garamond font files
test/run_tests.sh	Automated test runner